

DQ Profiling – Summary for Manager

What Was Done

A Snowflake stored procedure `PROFILE_TABLE_GENERIC` was built and run against Gold layer tables.

It captures and stores column-level metrics into `DQ_PROFILING.DQ_PROFILING_RESULTS`.

What Metrics Are Being Captured (Per Column)

Metric	What It Tells Us
<code>null_count</code>	How many nulls exist in the column
<code>distinct_count</code>	How many unique values
<code>duplicate_count_non_null</code>	Non-null values that repeat
<code>min_value / max_value</code>	Data range boundaries
<code>min_length / max_length / avg_length</code>	String length distribution
<code>dominant_pattern</code>	Most common format pattern (e.g. <code>nnnXnnX</code>)
<code>metric_value_pct</code>	Overall quality score (%)
<code>OBSERVATION</code>	e.g. "Duplicates identified"

What the Data Already Shows (From Sample Records)

- **null_count: 154,175 and 147,002** — very high nulls across key columns
- **duplicate_count: 25,807 and 20,879** — significant duplicates present
- **OBSERVATION = "Duplicates identified"** — on all 3 rows sampled

- **metric_value_pct: ~85–89%** — below 100%, meaning data quality issues exist
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What Is Currently Missing (The Gap)

The procedure **captures** metrics but does **not validate** them.

There are no rules that say:

- “null_count > X% = FAIL”
 - “duplicate_count > 0 = WARN”
 - “metric_value_pct < 80 = block pipeline”
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Proposed Actions to Leverage These Metrics

1. Define Thresholds → PASS / FAIL / WARN

Set business rules per metric and flag rows that breach them.

2. Trend Over Time Using RUN_TS

Compare run-over-run — detect if null count is growing month by month. This directly addresses Herney’s concern about DQ issues compounding.

3. Pipeline Integration

If a column FAILs threshold → either:

- Block promotion to Gold, OR
- Write to a DQ exceptions table for review

4. Business-Facing DQ Report / Dashboard

Use the `OBSERVATION` column + thresholds to produce a report the local team (Brazil) can review and act on.

Recommended Next Steps

Priority	Action	Owner
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High	Define threshold rules (null %, dup count)	Business + Data Team
High	Build validation layer on top of profiling results	Joshi
Medium	Trend analysis query using RUN_TS	Joshi
Medium	DQ exceptions table / dashboard	Joshi
Low	Extend profiling to Bronze/Silver layers	TBD

One-Line Manager Summary

"Profiling is running and metrics are captured. The next step is to define threshold rules so we can classify results as PASS/FAIL, trend them over time, and either block the pipeline or report issues to the business — starting with the high null and duplicate counts already visible in the Gold layer."